

Experiment 'p_2_8_rectangle_properties' Results

June 4, 2026

Experiment outcome: FAILURE_VERIFY

Bad responses: 1

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 5

Hard fails (resolution): 0

Soft fails (resolution): 1

Verification attempts: 4

Problem Specification

Problem name: p_2_8_rectangle_properties

Natural language statement: null

Method signature: p_2_8_rectangle_properties(length: real, width: real) returns (area: real, perimeter: real, diagonal: real)

Ensures

- $\text{area} == \text{length} * \text{width}$
- $\text{perimeter} == 2.0 * (\text{length} + \text{width})$
- $-0.01 < (\text{diagonal} * \text{diagonal}) - ((\text{length} * \text{length}) + (\text{width} * \text{width})) < 0.01$

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_2_8_rectangle_properties(length: real, width: real) returns (area: real, perimeter:
↪ real, diagonal: real)`

The method should respect the following contract:

`ensures area == length * width, ensures perimeter == 2.0 * (length + width), ensures
↪ -0.01 < (diagonal * diagonal) - ((length * length) + (width * width)) < 0.01`

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2nSYUgpWYvonnarnkGzY9jcyxjb

Model: gpt-5.5-2026-04-23

Created at: 1780581194

Finish reason: stop

Usage: 354 tokens in, and 1624 tokens out

```
{  
  area := length * width;  
  perimeter := 2.0 * (length + width);  
  diagonal :| -0.01 < (diagonal * diagonal) - ((length * length) + (width * width)) <  
    ↪ 0.01;  
}
```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

`problem_data/p_2_8_rectangle_properties/program.dfy(8,11): Error: cannot establish the
↪ existence of LHS values that satisfy the such-that predicate. Note, no trigger was
↪ found for the such-that predicate, which may be the reason the proof failed. To
↪ give a trigger explicitly, use the {:trigger} attribute. For more information, see
↪ the section on quantifier instantiation rules in the reference manual.`

```

8 | diagonal :| -0.01 < (diagonal * diagonal) - ((length * length) + (width * width)) <
    ↪ 0.01;
    | ^^

```

Dafny program verifier finished with 0 verified, 1 error

Correct the error in code by altering only the method body. Produce and show only the
 ↪ Dafny body, including the curly braces that surround it. Do not show the signature
 ↪ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2oqFHQ6AU87WwpS3A9XQBvUyrrb

Model: gpt-5.5-2026-04-23

Created at: 1780581280

Finish reason: stop

Usage: 696 tokens in, and 3731 tokens out

```

{
  area := length * width;
  perimeter := 2.0 * (length + width);

  var s := (length * length) + (width * width);
  assert 0.0 <= s;
  assert exists d: real :: d * d == s;
  diagonal :| diagonal * diagonal == s;
  assert (diagonal * diagonal) - s == 0.0;
}

```

Program → GenAI

When using dafny resolve, the below error is emitted and resolve fails:

```

problem_data/p_2_8_rectangle_properties/program.dfy(11,9): Warning: Could not find a
    ↪ trigger for this quantifier. Without a trigger, the quantifier may cause brittle
    ↪ verification. To silence this warning, add an explicit trigger using the {:trigger
    ↪ } attribute. For more information, see the section on quantifier instantiation
    ↪ rules in the reference manual.
    |
11 | assert exists d: real :: d * d == s;
    | ~~~~~

```

Compilation failed because warnings were found and --allow-warnings is false

Correct the error in code by altering only the method body. Produce and show only the
 ↪ Dafny body, including the curly braces that surround it. Do not show the signature
 ↪ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2qwpZ8AzAgJoxnkXt7Pbur7eDip

Model: gpt-5.5-2026-04-23

Created at: 1780581410

Finish reason: stop

Usage: 958 tokens in, and 4805 tokens out

```
{
  area := length * width;
  perimeter := 2.0 * (length + width);

  var s := (length * length) + (width * width);
  assert 0.0 <= s;
  assert exists d: real {:trigger d in {d}} :: d in {d} && d * d == s;
  diagonal :| diagonal in {diagonal} && diagonal * diagonal == s;
  assert (diagonal * diagonal) - s == 0.0;
  assert -0.01 < 0.0 < 0.01;
}
```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

```
Encountered internal compilation exception: Boogie program had 1 resolution errors:
Microsoft.Dafny.SourceOrigin: equality is not allowed in triggers
problem_data/p_2_8_rectangle_properties/program.dfy(1,7): Error: Internal error occurred
  ↳ during verification: Boogie program had 1 resolution errors:
Microsoft.Dafny.SourceOrigin: equality is not allowed in triggers
  at Microsoft.Boogie.ExecutionEngine.GetVerificationTasks(Program program,
    ↳ CancellationTokentoken cancellationToken)
  at Microsoft.Dafny.LanguageServer.Language.DafnyProgramVerifier.
    ↳ GetVerificationTasksAsync(ExecutionEngine engine, ResolutionResult resolution,
    ↳ ModuleDefinition moduleDefinition, CancellationTokentoken cancellationToken)
  at Microsoft.Dafny.Compilation.<>c__DisplayClass60_0.<<VerifyUnverifiedSymbol>b__1>d.
    ↳ MoveNext()
--- End of stack trace from previous location ---
  at Microsoft.Dafny.Compilation.VerifyUnverifiedSymbol(Boolean
    ↳ onlyPrepareVerificationForGutterTests, ICanVerify canVerify, ResolutionResult
    ↳ resolution, Func`2 taskFilter, Nullable`1 randomSeed)
  at DafnyDriver.Commands.CliCompilation.VerifyAllLazily(Nullable`1 randomSeed)+MoveNext
    ↳ ()
|
1 | method p_2_8_rectangle_properties(length: real, width: real) returns (area: real,
    ↳ perimeter: real, diagonal: real)
| ~~~~~
Unhandled exception. System.ArgumentException: Boogie program had 1 resolution errors:
Microsoft.Dafny.SourceOrigin: equality is not allowed in triggers
  at Microsoft.Boogie.ExecutionEngine.GetVerificationTasks(Program program,
    ↳ CancellationTokentoken cancellationToken)
```

```

at Microsoft.Dafny.LanguageServer.Language.DafnyProgramVerifier.
    ↳ GetVerificationTasksAsync(ExecutionEngine engine, ResolutionResult resolution,
    ↳ ModuleDefinition moduleDefinition, CancellationToken cancellationToken)
at Microsoft.Dafny.Compilation.<>c__DisplayClass60_0.<<VerifyUnverifiedSymbol>b__1>d.
    ↳ MoveNext()
--- End of stack trace from previous location ---
at Microsoft.Dafny.Compilation.VerifyUnverifiedSymbol(Boolean
    ↳ onlyPrepareVerificationForGutterTests, ICanVerify canVerify, ResolutionResult
    ↳ resolution, Func`2 taskFilter, Nullable`1 randomSeed)
at DafnyDriver.Commands.CliCompilation.VerifyAllLazily(Nullable`1 randomSeed)+MoveNext
    ↳ ()
at DafnyDriver.Commands.CliCompilation.VerifyAllLazily(Nullable`1 randomSeed)+System.
    ↳ Threading.Tasks.Sources.IValueTaskSource<System.Boolean>.GetResult()
at System.Linq.AsyncEnumerable.ToObservableObservable`1.<>c__DisplayClass2_0.<<
    ↳ Subscribe>g__Core|0>d.MoveNext() in /_/Ix.NET/Source/System.Linq.Async/System/
    ↳ Linq/Operators/ToObservable.cs:line 50
--- End of stack trace from previous location ---
at System.Reactive.PlatformServices.ExceptionServicesImpl.Rethrow(Exception exception)
    ↳ in /_/Rx.NET/Source/src/System.Reactive/Internal/ExceptionServicesImpl.cs:line
    ↳ 16
at System.Reactive.ExceptionHelpers.Throw(Exception exception) in /_/Rx.NET/Source/src
    ↳ /System.Reactive/Internal/ExceptionServices.cs:line 14
at System.Reactive.Stubs.<>c.<.cctor>b__2_1(Exception ex) in /_/Rx.NET/Source/src/
    ↳ System.Reactive/Internal/Stubs.cs:line 16
at System.Reactive.AnonymousObserver`1.OnErrorCore(Exception error) in /_/Rx.NET/
    ↳ Source/src/System.Reactive/AnonymousObserver.cs:line 73
at System.Reactive.ObserverBase`1.OnError(Exception error) in /_/Rx.NET/Source/src/
    ↳ System.Reactive/ObserverBase.cs:line 59
at System.Reactive.Subjects.Subject`1.OnError(Exception error) in /_/Rx.NET/Source/src
    ↳ /System.Reactive/Subjects/Subject.cs:line 103
at System.Linq.AsyncEnumerable.ToObservableObservable`1.<>c__DisplayClass2_0.<<
    ↳ Subscribe>g__Core|0>d.MoveNext() in /_/Ix.NET/Source/System.Linq.Async/System/
    ↳ Linq/Operators/ToObservable.cs:line 60
--- End of stack trace from previous location ---
at System.Linq.AsyncEnumerable.ToObservableObservable`1.<>c__DisplayClass2_0.<<
    ↳ Subscribe>g__Core|0>d.MoveNext() in /_/Ix.NET/Source/System.Linq.Async/System/
    ↳ Linq/Operators/ToObservable.cs:line 74
--- End of stack trace from previous location ---
at System.Threading.Tasks.Task.<>c.<ThrowAsync>b__128_1(Object state)
at System.Threading.ThreadPoolWorkQueue.Dispatch()
at System.Threading.PortableThreadPool.WorkerThread.WorkerThreadStart()

Correct the error in code by altering only the method body. Produce and show only the
    ↳ Dafny body, including the curly braces that surround it. Do not show the signature
    ↳ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

```

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2tluv3QsrkHMakY3BMaS4SDypKx

Model: gpt-5.5-2026-04-23

Created at: 1780581585

Finish reason: stop

Usage: 3011 tokens in, and 3798 tokens out

```
{
  area := length * width;
  perimeter := 2.0 * (length + width);

  var s := (length * length) + (width * width);
  assert 0.0 <= length * length;
  assert 0.0 <= width * width;
  assert 0.0 <= s;

  var triggerSet: set<real> := {};
  assert exists d: real { :trigger d in triggerSet } :: (d in triggerSet || !(d in
    ↪ triggerSet)) && d * d == s;
  diagonal :| (diagonal in triggerSet || !(diagonal in triggerSet)) && diagonal *
    ↪ diagonal == s;

  assert diagonal * diagonal == s;
  assert (diagonal * diagonal) - s == 0.0;
  assert -0.01 < 0.0 < 0.01;
}
```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

```
problem_data/p_2_8_rectangle_properties/program.dfy(15,2): Error: assertion might not
  ↪ hold
|
```

```
15 | assert exists d: real { :trigger d in triggerSet } :: (d in triggerSet || !(d in
  ↪ triggerSet)) && d * d == s;
| ~~~~~
```

```
problem_data/p_2_8_rectangle_properties/program.dfy(16,11): Error: cannot establish the
  ↪ existence of LHS values that satisfy the such-that predicate
|
```

```
16 | diagonal :| (diagonal in triggerSet || !(diagonal in triggerSet)) && diagonal *
  ↪ diagonal == s;
| ^^
```

Dafny program verifier finished with 0 verified, 2 errors

Correct the error in code by altering only the method body. Produce and show only the
↪ Dafny body, including the curly braces that surround it. Do not show the signature
↪ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2wBueZgTqYrGyXc6o4JsU42773E

Model: gpt-5.5-2026-04-23

Created at: 1780581735

Finish reason: stop

Usage: 2620 tokens in, and 1196 tokens out

```
{
  area := length * width;
  perimeter := 2.0 * (length + width);

  var s := length * length + width * width;
  assert 0.0 <= length * length;
  assert 0.0 <= width * width;
  assert 0.0 <= s;

  assert exists d: real {:trigger d * d} :: d * d == s;
  diagonal :| diagonal * diagonal == s;

  assert (diagonal * diagonal) - s == 0.0;
  assert -0.01 < 0.0;
  assert 0.0 < 0.01;
}
```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

```
problem_data/p_2_8_rectangle_properties/program.dfy(14,2): Error: assertion might not
  ↪ hold
  |
14 | assert exists d: real {:trigger d * d} :: d * d == s;
  | ~~~~~
```

Dafny program verifier finished with 0 verified, 1 error

Correct the error in code by altering only the method body. Produce and show only the
↪ Dafny body, including the curly braces that surround it. Do not show the signature
↪ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2xi0Zs53YifTELqDLaqA0sJpoWf

Model: gpt-5.5-2026-04-23

Created at: 1780581830

Finish reason: length

Usage: 2844 tokens in, and 7207 tokens out

The below is a raw response from GenAI that could not be parsed. It is likely that GenAI got 'stuck' in a loop and hit the pre-set token limit (i.e., finish reason is 'length').

Final Program

```
method p_2_8_rectangle_properties(length: real, width: real) returns (area: real,
    ↪ perimeter: real, diagonal: real)
    ensures area == length * width
    ensures perimeter == 2.0 * (length + width)
    ensures -0.01 < (diagonal * diagonal) - ((length * length) + (width * width)) <
        ↪ 0.01
{
    area := length * width;
    perimeter := 2.0 * (length + width);

    var s := length * length + width * width;
    assert 0.0 <= length * length;
    assert 0.0 <= width * width;
    assert 0.0 <= s;

    assert exists d: real {:trigger d * d} :: d * d == s;
    diagonal :| diagonal * diagonal == s;

    assert (diagonal * diagonal) - s == 0.0;
    assert -0.01 < 0.0;
    assert 0.0 < 0.01;
}
```

Total Token Usage

Input tokens: 10483

Output tokens: 22361

Reasoning tokens: 21514

Sum of 'total tokens': 32844

Experiment Timings

Overall Experiment started at 1780581192286, ended at 1780582016525, lasting 824239ms (824.24 seconds)

Iteration #4 started at 1780581582618, ended at 1780581734176, lasting 151558ms (151.56 seconds)

Iteration #5 started at 1780581734176, ended at 1780581823844, lasting 89668ms (89.67 seconds)

Iteration #6 started at 1780581823844, ended at 1780582016525, lasting 192681ms (192.68 seconds)

Iteration #1 started at 1780581192294, ended at 1780581279256, lasting 86962ms (86.96 seconds)

Iteration #2 started at 1780581279277, ended at 1780581408497, lasting 129220ms (129.22 seconds)

Iteration #3 started at 1780581408497, ended at 1780581582618, lasting 174121ms (174.12 seconds)

Experiment 'p_2_10_hybrid_car_cost' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_10_hybrid_car_cost

Natural language statement: null

Method signature: p_2_10_hybrid_car_cost(new_car_cost: real, miles_per_year: real, gas_price: real, mpg: real, resale_value: real) returns (total_cost: real)

Ensures

- `total_cost == new_car_cost + (miles_per_year / mpg) * gas_price * 5.0 - resale_value`

Requires

- `0.0 < mpg`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

```
p_2_10_hybrid_car_cost(new_car_cost: real, miles_per_year: real, gas_price: real, mpg:
    ↪ real, resale_value: real) returns (total_cost: real)
```

The method should respect the following contract:

```
requires 0.0 < mpg, ensures total_cost == new_car_cost + (miles_per_year / mpg) *
    ↪ gas_price * 5.0 - resale_value
```

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn30oCDZIJw64wgui48wUtd2XIPoy

Model: gpt-5.5-2026-04-23

Created at: 1780582022

Finish reason: stop

Usage: 350 tokens in, and 83 tokens out

```
{
  total_cost := new_car_cost + (miles_per_year / mpg) * gas_price * 5.0 - resale_value;
}
```

Final Program

```
method p_2_10_hybrid_car_cost(new_car_cost: real, miles_per_year: real, gas_price: real,
    ↪ mpg: real, resale_value: real) returns (total_cost: real)
  requires 0.0 < mpg
  ensures total_cost == new_car_cost + (miles_per_year / mpg) * gas_price * 5.0 -
    ↪ resale_value
{
  total_cost := new_car_cost + (miles_per_year / mpg) * gas_price * 5.0 - resale_value;
}
```

Total Token Usage

Input tokens: 350

Output tokens: 83

Reasoning tokens: 34

Sum of 'total tokens': 433

Experiment Timings

Overall Experiment started at 1780582021678, ended at 1780582058205, lasting 36527ms (36.53 seconds)

Iteration #1 started at 1780582021679, ended at 1780582058205, lasting 36526ms (36.53 seconds)

Experiment 'p_2_1_letter_dimensions' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_1_letter_dimensions

Natural language statement: null

Method signature: p_2_1_letter_dimensions() returns (width: real, height: real)

Ensures

- width == 8.5 * 25.4
- height == 11.0 * 25.4

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_2_1_letter_dimensions()` returns (width: real, height: real)

The method should respect the following contract:

`ensures width == 8.5 * 25.4, ensures height == 11.0 * 25.4`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2f6XKZM9hCQnUXbaHPDw2Q3bmRf

Model: gpt-5.5-2026-04-23

Created at: 1780580676

Finish reason: stop

Usage: 284 tokens in, and 211 tokens out

```
{  
  width := 8.5 * 25.4;  
  height := 11.0 * 25.4;  
}
```

Final Program

```
method p_2_1_letter_dimensions() returns (width: real, height: real)  
  ensures width == 8.5 * 25.4  
  ensures height == 11.0 * 25.4  
{  
  width := 8.5 * 25.4;  
  height := 11.0 * 25.4;  
}
```

Total Token Usage

Input tokens: 284

Output tokens: 211

Reasoning tokens: 163
Sum of ‘total tokens’: 495

Experiment Timings

Overall Experiment started at 1780580674149, ended at 1780580717971, lasting 43822ms (43.82 seconds)
Iteration #1 started at 1780580674224, ended at 1780580717969, lasting 43745ms (43.75 seconds)

Experiment ‘p_2_2_letter_perimeter’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_2_letter_perimeter

Natural language statement: null

Method signature: p_2_2_letter_perimeter() returns (perimeter: real, diagonal: real)

Ensures

- $\text{perimeter} == (8.5 + 11.0) * 2.0$
- $-0.01 < (\text{diagonal} * \text{diagonal}) - ((8.5 * 8.5) + (11.0 * 11.0)) < 0.01$

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_2_2_letter_perimeter()` returns (perimeter: real, diagonal: real)

The method should respect the following contract:

ensures perimeter == (8.5 + 11.0) * 2.0, ensures -0.01 < (diagonal * diagonal) - ((8.5 *
↪ 8.5) + (11.0 * 11.0)) < 0.01

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2fsE0DOSs1w7avoLDyLEryxliHK

Model: gpt-5.5-2026-04-23

Created at: 1780580724

Finish reason: stop

Usage: 354 tokens in, and 665 tokens out

```
{
  perimeter := (8.5 + 11.0) * 2.0;
  diagonal := 13.9015;
  assert diagonal * diagonal == 193.25170225;
  assert (8.5 * 8.5) + (11.0 * 11.0) == 193.25;
  assert (diagonal * diagonal) - ((8.5 * 8.5) + (11.0 * 11.0)) == 0.00170225;
  assert -0.01 < 0.00170225 < 0.01;
}
```

Final Program

```
method p_2_2_letter_perimeter() returns (perimeter: real, diagonal: real)
  ensures perimeter == (8.5 + 11.0) * 2.0
  ensures -0.01 < (diagonal * diagonal) - ((8.5 * 8.5) + (11.0 * 11.0)) < 0.01
{
  perimeter := (8.5 + 11.0) * 2.0;
  diagonal := 13.9015;
  assert diagonal * diagonal == 193.25170225;
```

```
assert (8.5 * 8.5) + (11.0 * 11.0) == 193.25;  
assert (diagonal * diagonal) - ((8.5 * 8.5) + (11.0 * 11.0)) == 0.00170225;  
assert -0.01 < 0.00170225 < 0.01;  
}
```

Total Token Usage

Input tokens: 354

Output tokens: 665

Reasoning tokens: 512

Sum of 'total tokens': 1019

Experiment Timings

Overall Experiment started at 1780580722636, ended at 1780580771285, lasting 48649ms (48.65 seconds)

Iteration #1 started at 1780580722685, ended at 1780580771261, lasting 48576ms (48.58 seconds)

Experiment ‘p_2_3_power_functions’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_3_power_functions

Natural language statement: null

Method signature: p_2_3_power_functions(x: int) returns (sq: int, cube: int, fourth: int)

Ensures

- `sq == x * x`
- `cube == x * x * x`
- `fourth == x * x * x * x`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_2_3_power_functions(x: int) returns (sq: int, cube: int, fourth: int)`

The method should respect the following contract:

`ensures sq == x * x, ensures cube == x * x * x, ensures fourth == x * x * x * x`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2gh2w7Vfkgqs1Mu8ENi0drmJSgL

Model: gpt-5.5-2026-04-23

Created at: 1780580775

Finish reason: stop

Usage: 300 tokens in, and 38 tokens out

```
{
  sq := x * x;
  cube := sq * x;
  fourth := sq * sq;
}
```

Final Program

```
method p_2_3_power_functions(x: int) returns (sq: int, cube: int, fourth: int)
  ensures sq == x * x
  ensures cube == x * x * x
  ensures fourth == x * x * x * x
{
  sq := x * x;
  cube := sq * x;
  fourth := sq * sq;
}
```

Total Token Usage

Input tokens: 300

Output tokens: 38

Reasoning tokens: 0

Sum of 'total tokens': 338

Experiment Timings

Overall Experiment started at 1780580773209, ended at 1780580799043, lasting 25834ms (25.83 seconds)

Iteration #1 started at 1780580773214, ended at 1780580799043, lasting 25829ms (25.83 seconds)

Experiment 'p_2_4_a_sum' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_4_a_sum

Natural language statement: null

Method signature: p_2_4_a_sum(x: int, y: int) returns (result: int)

Ensures

- `result == x + y`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_2_4_a_sum(x: int, y: int) returns (result: int)`

The method should respect the following contract:

`ensures result == x + y`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2h6rhSdLaCTAguRm58E3xeet6kI

Model: gpt-5.5-2026-04-23

Created at: 1780580800

Finish reason: stop

Usage: 252 tokens in, and 63 tokens out

```
{  
  result := x + y;  
}
```

Final Program

```
method p_2_4_a_sum(x: int, y: int) returns (result: int)  
  ensures result == x + y  
{  
  result := x + y;  
}
```

Total Token Usage

Input tokens: 252

Output tokens: 63

Reasoning tokens: 35

Sum of ‘total tokens’: 315

Experiment Timings

Overall Experiment started at 1780580800461, ended at 1780580828783, lasting 28322ms (28.32 seconds)

Iteration #1 started at 1780580800463, ended at 1780580828783, lasting 28320ms (28.32 seconds)

Experiment 'p_2_4_b_difference' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_4_b_difference

Natural language statement: null

Method signature: p_2_4_b_difference(x: int, y: int) returns (result: int)

Ensures

- `result == x - y`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_2_4_b_difference(x: int, y: int) returns (result: int)`

The method should respect the following contract:

`ensures result == x - y`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2hcUFS8SjlJ5IKJvTWNDZrGGCHn

Model: gpt-5.5-2026-04-23

Created at: 1780580832

Finish reason: stop

Usage: 252 tokens in, and 43 tokens out

```
{  
  result := x - y;  
}
```

Final Program

```
method p_2_4_b_difference(x: int, y: int) returns (result: int)  
  ensures result == x - y  
{  
  result := x - y;  
}
```

Total Token Usage

Input tokens: 252

Output tokens: 43

Reasoning tokens: 15

Sum of ‘total tokens’: 295

Experiment Timings

Overall Experiment started at 1780580832094, ended at 1780580856200, lasting 24106ms (24.11 seconds)

Iteration #1 started at 1780580832096, ended at 1780580856199, lasting 24103ms (24.10 seconds)

Experiment 'p_2_4_c_product' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_4_c_product

Natural language statement: null

Method signature: p_2_4_c_product(x: int, y: int) returns (result: int)

Ensures

- `result == x * y`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_2_4_c_product(x: int, y: int) returns (result: int)`

The method should respect the following contract:

`ensures result == x * y`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2i1V6Vq3HjmvRykTPz8VNnIyCpk

Model: gpt-5.5-2026-04-23

Created at: 1780580857

Finish reason: stop

Usage: 252 tokens in, and 55 tokens out

```
{  
  result := x * y;  
}
```

Final Program

```
method p_2_4_c_product(x: int, y: int) returns (result: int)  
  ensures result == x * y  
{  
  result := x * y;  
}
```

Total Token Usage

Input tokens: 252

Output tokens: 55

Reasoning tokens: 27

Sum of ‘total tokens’: 307

Experiment Timings

Overall Experiment started at 1780580857236, ended at 1780580913457, lasting 56221ms (56.22 seconds)

Iteration #1 started at 1780580857246, ended at 1780580913451, lasting 56205ms (56.21 seconds)

Experiment 'p_2_4_d_avarage' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_4_d_avarage

Natural language statement: null

Method signature: p_2_4_d_avarage(x: int, y: int) returns (result: real)

Ensures

- `result == (x + y) as real / 2.0`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_2_4_d_avarage(x: int, y: int) returns (result: real)`

The method should respect the following contract:

`ensures result == (x + y) as real / 2.0`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2j41CJifHOawR9eqiFCLwOTF8py

Model: gpt-5.5-2026-04-23

Created at: 1780580922

Finish reason: stop

Usage: 274 tokens in, and 150 tokens out

```
{  
  result := (x + y) as real / 2.0;  
}
```

Final Program

```
method p_2_4_d_avarage(x: int, y: int) returns (result: real)  
  ensures result == (x + y) as real / 2.0  
{  
  result := (x + y) as real / 2.0;  
}
```

Total Token Usage

Input tokens: 274

Output tokens: 150

Reasoning tokens: 113

Sum of ‘total tokens’: 424

Experiment Timings

Overall Experiment started at 1780580921519, ended at 1780580962848, lasting 41329ms (41.33 seconds)

Iteration #1 started at 1780580921520, ended at 1780580962848, lasting 41328ms (41.33 seconds)

Experiment 'p_2_4_e_distance' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_4_e_distance

Natural language statement: null

Method signature: p_2_4_e_distance(x: int, y: int) returns (result: int)

Ensures

- `result >= 0`
- `if x >= y then result == x - y else result == y - x`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_2_4_e_distance(x: int, y: int) returns (result: int)`

The method should respect the following contract:

`ensures result >= 0, ensures if x >= y then result == x - y else result == y - x`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2jo2lO9Em7iZD2hdE1P2vNKuDr7

Model: gpt-5.5-2026-04-23

Created at: 1780580968

Finish reason: stop

Usage: 286 tokens in, and 158 tokens out

```
{
  if x >= y {
    result := x - y;
  } else {
    result := y - x;
  }
}
```

Final Program

```
method p_2_4_e_distance(x: int, y: int) returns (result: int)
  ensures result >= 0
  ensures if x >= y then result == x - y else result == y - x
{
  if x >= y {
    result := x - y;
  } else {
    result := y - x;
  }
}
```

}

Total Token Usage

Input tokens: 286

Output tokens: 158

Reasoning tokens: 107

Sum of 'total tokens': 444

Experiment Timings

Overall Experiment started at 1780580966160, ended at 1780581002473, lasting 36313ms (36.31 seconds)

Iteration #1 started at 1780580966162, ended at 1780581002473, lasting 36311ms (36.31 seconds)

Experiment 'p_2_4_f_max' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_4_f_max

Natural language statement: null

Method signature: p_2_4_f_max(x: int, y: int) returns (result: int)

Ensures

- `result == x || result == y`
- `if x >= y then result == x else result == y`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_2_4_f_max(x: int, y: int) returns (result: int)`

The method should respect the following contract:

`ensures result == x || result == y, ensures if x >= y then result == x else result == y`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2kQkDuZcAinZ3VCM45MpIIY0PyH

Model: gpt-5.5-2026-04-23

Created at: 1780581006

Finish reason: stop

Usage: 284 tokens in, and 41 tokens out

```
{
  if x >= y {
    result := x;
  } else {
    result := y;
  }
}
```

Final Program

```
method p_2_4_f_max(x: int, y: int) returns (result: int)
  ensures result == x || result == y
  ensures if x >= y then result == x else result == y
{
  if x >= y {
    result := x;
  } else {
    result := y;
  }
}
```

}

Total Token Usage

Input tokens: 284

Output tokens: 41

Reasoning tokens: 0

Sum of 'total tokens': 325

Experiment Timings

Overall Experiment started at 1780581005176, ended at 1780581038548, lasting 33372ms (33.37 seconds)

Iteration #1 started at 1780581005182, ended at 1780581038547, lasting 33365ms (33.37 seconds)

Experiment 'p_2_4_g_min' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_4_g_min

Natural language statement: null

Method signature: p_2_4_g_min(x: int, y: int) returns (result: int)

Ensures

- `result == x || result == y`
- `if x <= y then result == x else result == y`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_2_4_g_min(x: int, y: int) returns (result: int)`

The method should respect the following contract:

`ensures result == x || result == y, ensures if x <= y then result == x else result == y`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2l3N6GyzX6P2jyquJvPADA9APts

Model: gpt-5.5-2026-04-23

Created at: 1780581045

Finish reason: stop

Usage: 284 tokens in, and 72 tokens out

```
{
  if x <= y {
    result := x;
  } else {
    result := y;
  }
}
```

Final Program

```
method p_2_4_g_min(x: int, y: int) returns (result: int)
  ensures result == x || result == y
  ensures if x <= y then result == x else result == y
{
  if x <= y {
    result := x;
  } else {
    result := y;
  }
}
```

}

Total Token Usage

Input tokens: 284

Output tokens: 72

Reasoning tokens: 25

Sum of 'total tokens': 356

Experiment Timings

Overall Experiment started at 1780581044020, ended at 1780581109084, lasting 65064ms (65.06 seconds)

Iteration #1 started at 1780581044025, ended at 1780581109084, lasting 65059ms (65.06 seconds)

Experiment ‘p_2_6_convert’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_6_convert

Natural language statement: null

Method signature: p_2_6_convert(x: real) returns (miles: real, feet: real, inches: real)

Ensures

- `x / 0.000621371 == miles`
- `x / 3.28084 == feet`
- `x / 39.3701 == inches`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_2_6_convert(x: real) returns (miles: real, feet: real, inches: real)`

The method should respect the following contract:

`ensures x / 0.000621371 == miles, ensures x / 3.28084 == feet, ensures x / 39.3701 ==`
→ `inches`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2mA5WCaU7Jz2M2JIY1yeZyFUqBr

Model: gpt-5.5-2026-04-23

Created at: 1780581114

Finish reason: stop

Usage: 314 tokens in, and 186 tokens out

```
{
  miles := x / 0.000621371;
  feet := x / 3.28084;
  inches := x / 39.3701;
}
```

Final Program

```
method p_2_6_convert(x: real) returns (miles: real, feet: real, inches: real)
  ensures x / 0.000621371 == miles
  ensures x / 3.28084 == feet
  ensures x / 39.3701 == inches
{
  miles := x / 0.000621371;
  feet := x / 3.28084;
  inches := x / 39.3701;
}
```

Total Token Usage

Input tokens: 314

Output tokens: 186

Reasoning tokens: 129

Sum of 'total tokens': 500

Experiment Timings

Overall Experiment started at 1780581111850, ended at 1780581153047, lasting 41197ms (41.20 seconds)

Iteration #1 started at 1780581111851, ended at 1780581153047, lasting 41196ms (41.20 seconds)

Experiment 'p_2_7_circle_properties' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_7_circle_properties

Natural language statement: null

Method signature: p_2_7_circle_properties(radius: real) returns (circumference: real, area: real, volume: real, surface_area: real)

Ensures

- `circumference == 2.0 * 3.14159 * radius`
- `area == 3.14159 * radius * radius`
- `volume == (4.0 / 3.0) * 3.14159 * radius * radius * radius`
- `surface_area == 4.0 * 3.14159 * radius * radius`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_2_7_circle_properties(radius: real)` returns `(circumference: real, area: real, volume: real, surface_area: real)`

The method should respect the following contract:

`ensures circumference == 2.0 * 3.14159 * radius, ensures area == 3.14159 * radius * radius, ensures volume == (4.0 / 3.0) * 3.14159 * radius * radius * radius, ensures surface_area == 4.0 * 3.14159 * radius * radius`

Produce and show only the Dafny body of this method, including the curly braces that surround it. Do not show the signature nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2mr64ILOU3NnNSeSyRSxylFQ5m8

Model: gpt-5.5-2026-04-23

Created at: 1780581157

Finish reason: stop

Usage: 408 tokens in, and 200 tokens out

```
{
  circumference := 2.0 * 3.14159 * radius;
  area := 3.14159 * radius * radius;
  volume := (4.0 / 3.0) * 3.14159 * radius * radius * radius;
  surface_area := 4.0 * 3.14159 * radius * radius;
}
```

Final Program

```
method p_2_7_circle_properties(radius: real) returns (circumference: real, area: real,
  volume: real, surface_area: real)
  ensures circumference == 2.0 * 3.14159 * radius
  ensures area == 3.14159 * radius * radius
  ensures volume == (4.0 / 3.0) * 3.14159 * radius * radius * radius
  ensures surface_area == 4.0 * 3.14159 * radius * radius
{
```

```
circumference := 2.0 * 3.14159 * radius;  
area := 3.14159 * radius * radius;  
volume := (4.0 / 3.0) * 3.14159 * radius * radius * radius;  
surface_area := 4.0 * 3.14159 * radius * radius;  
}
```

Total Token Usage

Input tokens: 408

Output tokens: 200

Reasoning tokens: 102

Sum of 'total tokens': 608

Experiment Timings

Overall Experiment started at 1780581156299, ended at 1780581191131, lasting 34832ms (34.83 seconds)

Iteration #1 started at 1780581156305, ended at 1780581191131, lasting 34826ms (34.83 seconds)

